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RESEARCH ARTICLES

LATE MESOZOIC PALAEOMAGNETISM FROM SPITSBERGEN; IMPLICATIONS FOR CONTINENTAL DRIFT IN THE ARCTIC *

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Thermal and alternating-current demagnetization combined with ore microscopy and measurements of the temperature dependence of saturation magnetization have been carried out on some Mesozoic, probably Cretaceous, basaltic lavas from two areas (Seidfjell and Sørlifjell) at Spitsbergen. The experimental studies suggest that the Seidfjell locality has undergone extensive oxidations, which resulted in remagnetization. The estimated palaeomagnetic pole for this area is 77°N 107°E, which suggests a remagnetization, probably some time in the Late Tertiary. On the other hand the experimental data from the Sørlifjell locality suggest that the magnetization is primarily of deuteric origin. The mean palaeomagnetic pole position for this latter formation is at 75°N 235°E, which is significantly different from previously published European Mesozoic data. However, closing the Neo-Arctic basin by rotating Spitsbergen towards the Lomonosov Ridge, makes the suggested Cretaceous pole coincide with poles of similar age from North-America. This suggests that the estimated pole from Sørlifjell is a good approximation for a Late Mesozoic palaeomagnetic pole for Europe and it also confirms that the process of continental separation in the Arctic has taken place in Tertiary time.

1. Introduction

Palaeomagnetism of the Mesozoic of Europe is highly unexplored. To a large extent this is due to unsuitable rocks; most results being based on poorly consolidated sediments which are likely to suffer from the remagnetization complexity. The Svalbard Archipelago may to some extent help in solving this problem because of the considerable amount of Jurassic and Cretaceous volcanism of this region. On the other hand, previous palaeomagnetic studies of the Mesozoic of Svalbard (Krumsiek et al., 1968; Spall, 1968) have given strongly diverging results, suggesting that remagnetization plays an important role in these rocks as well. This remagnetization hypothesis is recently confirmed by Halvorsen (1973).

Svalbard is situated at the Barents Shelf, which is taken as the northwestern corner of the European continent, and at least since post-Devonian time there appears to have been no relative movement between the Svalbard region and the major European continent (Storetvedt, 1972). Thus, the palaeomagnetic investigation of the Mesozoic lavas at Svalbard will possibly contribute to elucidate the Jurassic–Cretaceous palaeomagnetic field relative to Europe. Furthermore, the data will be used in adding information on the evolution history in the Arctic.

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2. Geology and sampling

Spitsbergen, which is the main island in the archipelago termed Svalbard, is situated in the Barents Sea between 76–80°N and 10–22°E. The lava sequence is found between Woodfjord and Wijdefjord, forming caps on the top of the mountains. They are considered as the remainings of an earlier lava plateau. The preserved sequence consists of at least eighteen horizontal flows, the thickness of each ranging from 1 to 40 m. The lavas have been classified as plagioclase basalts (Goldschmidt, 1911; Hoel and Holtedahl, 1911; Hoel, 1914).

According to Orvin (1940) the lavas have been extruded on an eroded surface of red and grey sandstone, which has been referred to as the Cretaceous

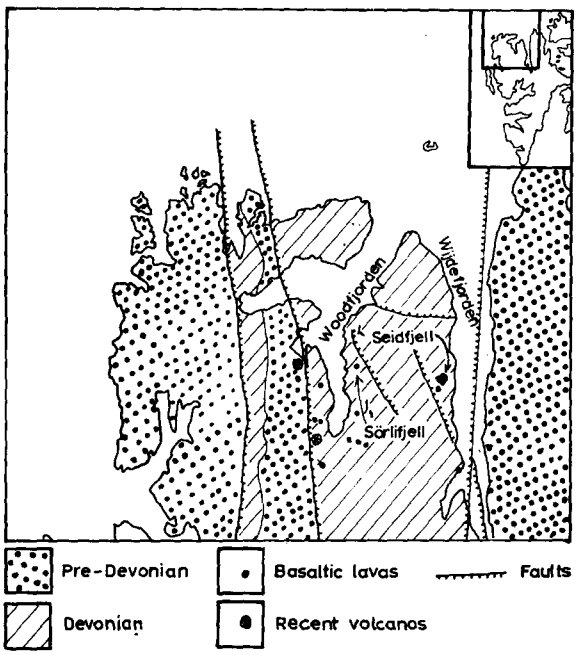


Fig.1. Geological sketch map showing the two areas sampled (Seidfjell and Sørlifjell).

peneplain. The lavas have not been subjected to radiometric age determinations, but stratigraphic age relations indicate either a Mesozoic or a Cenozoic age.

Tyrrell and Sandford (1933) suggest that the basaltic lavas are the effusive equivalents of the dolerite intrusions, which based on radiometric dating (Gayer et al., 1966) are thought to be of Late Jurassic age (149 ±11 m.y.). On the other hand, Harland (1969) has suggested that the lavas date back to the Late Cretaceous or possibly to the Early Tertiary; extruded in connection with the opening of the Arctic–North Atlantic Ocean basin.

Apart from the western Spitsbergen orogeny, which is considered to have resulted from a northward movement of Greenland (in the Early Tertiary), the major part of Spitsbergen seems to have been part of a relatively stable crust in the Mesozoic as well as in the Cenozoic.

The rock collection, which has been taken in two different areas (cf. Fig.1), comprises 63 oriented hand samples from 16 different sites. 12 sites (47 samples) originate from Sørlifjell and the remaining 4 sites (16 samples) from Seidfjell.

3. The magnetic minerals

Polished sections from 5 sites (8 samples) have been subjected to ore microscopy. In order to distinguish between strongly magnetic and weakly magnetic (or non-magnetic) phases the colloid technique has been applied. Furthermore, the temperature dependence of saturation magnetization, J_s, has been measured with a translation balance. The heatings were carried out in air in applied field strengths varying between 5000 and 6500 Oe.

The magneto-mineralogical results are consistent covering the whole stratigraphic sequence in both areas, and they can be summarized as follows:

3.1. Seidfjell

The iron–titanium oxides have been strongly altered and no traces of the primary titanomagnetite has been recognized. The ilmenite grains often show intergrowths of haematite and other alteration products. Furthermore, the olivine grains have undergone serpentinization. This latter process appears to have resulted in the formation of a mixture of magnetite, haematite, goethite, martite and probably maghemite (cation-deficient titanomagnetite). The altered olivine grains attract the magnetic colloid indicating the presence of magnetite and/or maghemite. Parts of the ilmenite grains are strongly magnetic as well. The magnetic grains have an estimated effective grain size $< 5\ \mu$.

The saturation magnetization, J_s, shows that all samples examined have a Curie temperature, T_c, in the region 570–600°C. The heating and cooling curves are always irreversible and all samples show a decrease of J_s during cooling, together with a systematic lowering of the T_c by about 30–60°C. Curie temperatures in the range of 570–600°C may be associated with magnetite, haemo-ilmenite or chemically stabilized titanomaghemite (cation-deficient titanomagnetites). Owing to the particular decrease of J_s on heating it is possible that, besides true Curie temperatures, one is dealing with transformation temperatures of titanomaghemite as well. The presence of haematite observed in the microscope is not confirmed by the J_s–T measurements, and the reason for this is obviously the low specific magnetization of this mineral. On a whole the primary ore minerals of the Seidfjell

'locality' seem to have been strongly altered by low-temperature processes.

3.2. Sørlifjell

The ore minerals of these samples seem to consist of optically homogeneous titanomagnetite grains. The quantity of ilmenite varies from sample to sample and no traces of exsolved ilmenite lamellae appear to exist. Neither the olivine nor the iron–titanium oxides show signs of alteration. Furthermore, the magnetic colloid technique shows that the quantity of more strongly magnetic mineral phases of the opaques varies from sample to sample. The magnetic grains have an effective grain size in the range of 5–50 μ.

J_s–T measurements show that all samples have a single Curie temperature, T_c, in the region 180–340°C. Furthermore, the heating and cooling curves are always irreversible. Two samples showed a minor increase in the J_s during heating at about 450°C.

According to Buddington and Lindsley (1964) and Carmichal and Nicholls (1967) the observed T_c-range corresponds roughly to the composition of the virgin-state titanomagnetites in basaltic rocks.

After heating to 700°C the T_c increases to approximately 550°C while the J_s increases by 20–50% (measured at room temperature). The minor increase in the J_s during heating, together with the fact that all cooling curves are above the heating curves, suggests exolution-oxidation together with unmixing processes in cation-deficient spinel phases as the ore mineral alteration during heating.

3.3. Conclusions

The magneto-mineralogical examination reveals that the rocks from the two areas have undergone different alterations; rocks from Seidfjell being strongly oxidized while rocks from Sørlifjell only show feeble low-temperature oxidation. The Curie temperatures of 600°C (Seidfjell) are slightly above the expected T_c for pure magnetite indicating the presence of cation-deficient spinel phases.

Furthermore, the T_c of 340 (Sørlifjell) is also above what is normally expected in basaltic titanomagnetite, indicating that low-temperature oxidation may be of importance in these rocks as well. According to Readman and O'Reilly (1971) the minor increase in J_s during heating (at about 450°C) may represent an unmixing process in cation-deficient spinels. Whether this latter process is a result of oxidation during the heating or represents some earlier low-temperature oxidation is impossible to say.

4. Demagnetization experiments

In order to test the stability of the natural remanent magnetization (N.R.M.) and to remove relatively softer secondary components of magnetization, a large number of specimens were subjected to progressive alternating-current (a.c.) and thermal demagnetization. The measurements of the remanent magnetization have been carried out on two different types of magnetometers. The intensity of N.R.M. is distributed between the orders of 10^{-2} e.m.u./cm^3 and 10^{-4} e.m.u./cm^3.

4.1. A.c.-demagnetization results

A total of 63 specimens (one from each hand sample) have been subjected to a.c.-demagnetization and with no exception all specimens from Seidfjell are more resistant against a.c. demagnetization than the specimens from Sørlifjell. Recalling the difference in state of oxidation of these rocks the observed difference in magnetic hardness is as expected (Larson et al., 1969). Furthermore, low-stability viscous mag-

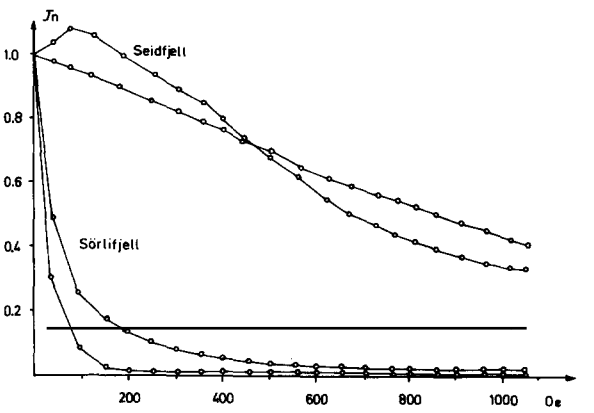


Fig.2. Examples of remanence intensity versus alternating field illustrating the magnetic stability difference between Seidfjell and Sørlifjell.

netization is of no importance in the specimens from Seidfjell while it is of some importance in the specimens from Sørlifjell. All the specimens from Seidfjell have a characteristic fossil magnetization which is directed easterly with a fairly steep downward (positive) inclination and the directions remain unchanged even when subjected to a.c.-demagnetization in excess of 1000 Oe. On the other hand, in some of the specimens from Sørlifjell the magnetic vector shows more or less systematic movements indicating a removal of some unstable magnetization. This latter magnetization is entirely removed by a.c.-demagnetization in about 150 Oe and the characteristic stable fossil magnetization is directed northwesterly with a fairly steep downward (positive) inclination. Characteristic examples of intensity behaviour during progressive a.c.-demagnetization are shown in Fig.2.

4.2. Thermal results

A total of 63 specimens (one from each hand sample) have been subjected to thermal demagnetization as well. First of all there is a marked contrast in thermal stability, the remanence intensity of the specimens from Sørlifjell being commonly reduced by about 60–90% after heating to 300°C while the specimens from Seidfjell had to be thermally demagnetized to about 550°C to achieve a similar intensity reduction. This confirms the difference in oxidation-state of the rocks from the two areas considered.

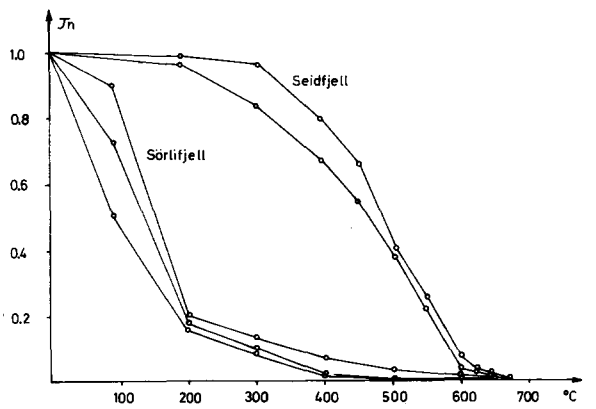


Fig.3. Examples of remanence intensity versus temperature illustrating the magnetic stability difference between Seidfjell and Sørlifjell.

Examples of this difference in the decay of remanence intensity as a function of temperature are shown in Fig.3.

It appears reasonable to conclude that the dominant remanence-carrying minerals in the Seidfjell lavas are magnetite and haematite, while a "virgin"-state titanomagnetite phase contributes to the essential remanence intensity of the Sørlifjell lava. This observation agrees with the ore microscopy data and with the saturation magnetization measurements.

In the majority of specimens the direction of N.R.M. has a steep downward (positive) inclination, but when subjected to thermal demagnetization the specimens from Seidfjell and Sørlifjell show a different directional behaviour. All specimens from Seidfjell have a stable fossil magnetization directed easterly showing no directional movement on demagnetization to temperatures of approximately 640°C. On subsequent demagnetization to 670°C the magnetic vector shows in general irregular directional movement. At this stage the intensity of the remanence varies between 8.10^{-7} and 3.10^{5} e.m.u./cm^3 (less than 0.3% of the original remanence). The directions are random and consequently they are set aside in the further consideration.

On the other hand, most specimens from Sørlifjell have a stable normal magnetization direction directed northwesterly, showing no directional movements when demagnetized to temperatures of approximately 300°C. A few specimens show a systematic directional movement towards the magnetization of this "locality" when being demagnetized to about 300–400°C. In most cases the direction is stable beyond the major blocking temperature spectrum, carrying in some cases more than 90% of the remanence intensity. Furthermore, on demagnetization to higher temperatures the magnetic vector initiates irregular movements. According to previous considerations this latter stage has no geophysical relevance and the associated directions have been omitted from further discussion.

5. A Mesozoic pole for Europe

The magneto-mineralogical examination together with demagnetization results reveal that the specimens from Seidfjell are relatively much more oxi-

dized than the Sørlifjell ones. The oxidation seems dominantly to be of the low-temperature type.

From the directional behaviour during demagnetization, the specimens from Seidfjell show a stable magnetization direction when demagnetized to 640°C or 1000 Oe. Thus, it seems likely to suggest that the low-temperature chemical magnetization (C.R.M.) originated in a fixed dipole field; the time span for C.R.M. acquisition represents no relative polar migration. The experimental data of specimens from Sørlifjell indicate the presence of stable primary T.R.M., possibly together with feeble C.R.M. The stability of some of the directions beyond the major blocking temperature spectrum reveals that minor C.R.M. components may be present and it also indicates that this C.R.M. magnetization originated in the same dipole field as the T.R.M.

The distribution of bulk magnetization direction is given in Fig.4. These latter directions are sample means, each being the mean of two bulk specimen directions (except for some scattered results previously discussed). The bulk directions clearly reveal a significant difference in the magnetization direction from Seidfjell and Sørlifjell; the specimens from Seidfjell having an easterly direction, while the specimens from Sørlifjell have a northwesterly one.

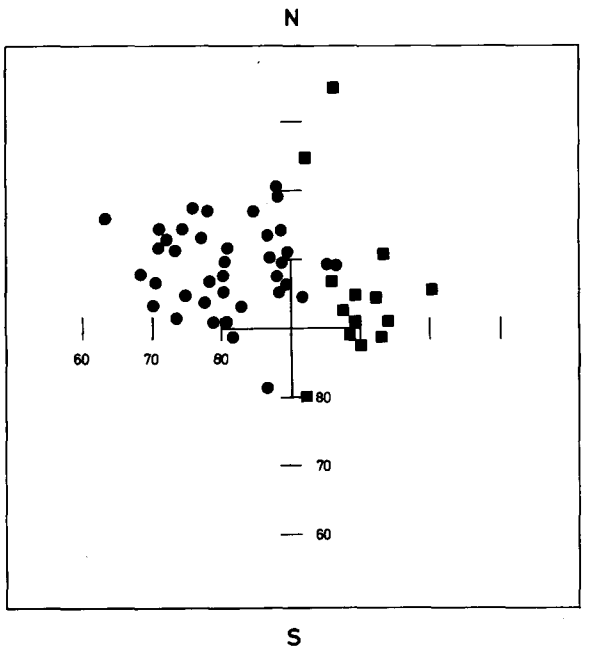


Fig. 4. Sample mean directions of stable bulk magnetization. Circles represent Sørlifjell, squares Seidfjell.

In order to make palaeomagnetic correlation, two palaeomagnetic pole positions were calculated, one from each of the two areas considered. There may be some doubt as to the statistical significance of the data and for this reason the mean pole is calculated with unit weight on specimen, sample and site respectively. The mean poles are listed in Table I together with a Jurassic pole position calculated by Briseid (1972). (Statistical parameters are according to Fisher, 1953). It is apparent from the table that the mean poles with unit weight on specimen, sample and site, all overlap one another, suggesting that the choice of unit weight is statistically insignificant. However, it appears likely that the mean pole calculated with unit weight on site is the most representative choice because it probably is least biased by effects of secular variation. Furthermore, Table I shows that the mean poles from Seidfjell, Sørlifjell and the Jurassic dolerite intrusion all are significantly different, and this means that they have achieved their stable magnetization direction at different times. With reference to our previous reasoning it seems likely to conclude that the lavas from Sørlifjell reflect the geomagnetic field at the time the igneous activity took place, while the lavas from Seidfjell most likely have been remagnetized in some later period. However, we cannot completely ignore the possibility that the lavas at Seidfjell represent some volcanic events postdating the lavas at Sørlifjell. The difference in the magnetization direction of the Sørlifjell lavas and the Jurassic dolerite intrusion indicate that the basaltic lavas and the dolerites are from two different geological periods. Thus, we conclude that the lava plateau is of post-Jurassic age, probably Late Cretaceous, and that the calculated mean pole from Sørlifjell represents the Late Cretaceous palaeomagnetic pole for Spitsbergen. This latter pole differs significantly from other European Cretaceous poles previously published (Costa-Foru et al., 1964; Bucha, 1965). Comparison with other recently available Mesozoic data from Svalbard (Halvorsen, 1973) suggests a more easterly path for the polar wandering curve of northern Europe; strengthening the above conclusion that the Sørlifjell result is a good approximation for a Late Mesozoic palaeomagnetic pole for Europe. The mean pole from Seidfjell is located in the same area as the major part of the Tertiary palaeomagnetic poles. Thus, we conclude that the

TABLE I
Palaeomagnetic results from the Spitsbergen basalts and dolerites

Sample number	K	α_{95}	D_m	I_m	Pole	Unit weight on
Sørlifjell						
92	27.11	2.88	334	77	75°N 237°E	bulk specimen direction
47	35.32	3.55	333	77	75°N 237°E	sample mean
12	109.80	4.16	335	77	75°N 235°E	site mean
Seidfjell						
32	44.75	3.84	48	81	77°N 107°E	bulk specimen direction
16	44.71	5.58	48	81	77°N 107°E	sample mean
4	44.73	6.94	48	81	77°N 107°E	site mean
Jurassic dolerite						
38	95.9	3.00	171	−68	61°N 210°E	bulk specimen direction (Briseid, 1972)

magnetization from Seidfjell is of post-Cretaceous age. Whether this mean pole represents a true palaeomagnetic pole or a spot reading is impossible to say, but it appears reasonable to believe that the time span of C.R.M. acquisition represents at least the period of secular variation (10^4–10^5 year).

In order to get light on the dating of the remagnetization of the Seidfjell lavas, further investigations are essential.

6. The opening of the Neo-Arctic

According to Harland (1969) the Arctic Ocean consists of two ocean basins; the Neo-Arctic and the Paleo-Arctic Ocean basins which are separated by the Lomonosov Ridge. The origin of the Atlantic and Arctic Ocean in Mesozoic and Cenozoic times by fission and growth along "mid oceanic" rifts is now a familiar concept. Wilson (1963) has argued that the Lomonosov Ridge must be regarded as a former part of the northern edge of the Barents Shelf. It has been suggested (Harland, 1969) that the Paleo-Arctic basin has existed throughout Phanerozoic time, while the Neo-Arctic basin has existed for only about the last 30 m.y. On the other hand, Ostenso (1972) assumes that the Paleo-Arctic Ocean began opening some 180 m.y. ago with the so-called Alpha Ridge as spreading axis until 40 m.y. ago when the spreading axis changed. The new axis of arctic spreading intersected the Eurasia continental margin, slicing off a section which was translated northward to form the present Lomonosov Ridge, creating the presently active Nansen Ridge (situated in the middle of the Neo-Arctic basin). Thus, if the opening of the Neo-Arctic basin is a Tertiary event and probably post-dating the magnetization of the Sørlifjell lavas, the considered palaeomagnetic pole positions for the Late Mesozoic of Spitsbergen should be significantly different from

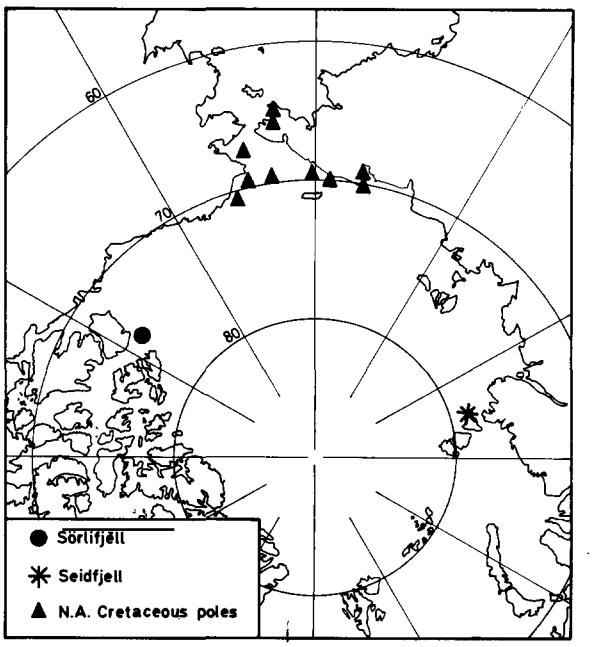


Fig.5. Palaeomagnetic poles from Sørlifjell and Seidfjell together with North-American Cretaceous poles.

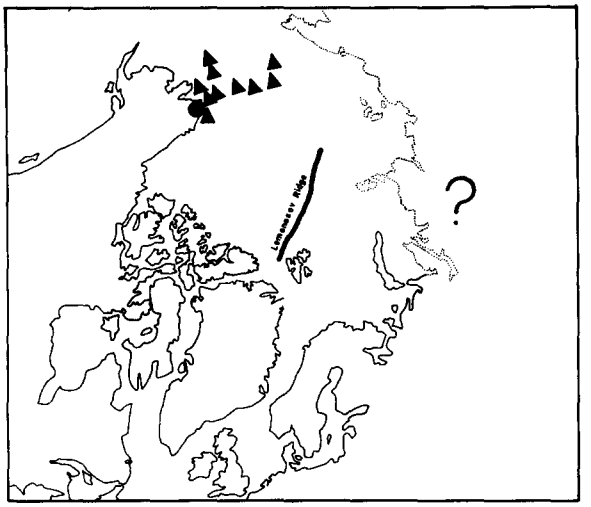


Fig.6.The continental configuration after the reconstruction in the Arctic. Further explanation in the text. (Symbols: See Fig.5.)

Late Cretaceous poles for North-America. Fig.5 shows the palaeomagnetic poles of this paper together with poles of similar age from North-America. A closing-up of the Neo-Arctic basin by moving Spitsbergen northwards against the Lomonosov Ridge ought to give a better accordance between the Cretaceous poles for North-America and the Spitsbergen one. Le Pichon (1968) has suggested that the opening-up of the Arctic–North Atlantic Ocean took place as a rotation with a pole of opening at 78°N 102°E. This estimate is based on four fracture zones outside the Neo-Arctic basin. In an attempt to reconstruct the Arctic palaeogeography prior to the evolution of the Neo-Arctic basin, Spitsbergen and the European continent have been rotated northward to the Lomonosov Ridge using this pole of opening. The actual relative movement, however, may well have been much more complex than that of a simple rotation. The reconstruction of the Arctic palaeogeography is shown in Fig.6. This figure also shows the Cretaceous poles from North-America together with the rotated Sørlifjell pole. (Greenland is replaced to its supposed Late Cretaceous position in accordance with Harland (1969).) Fig.5 and 6 show that, before the reconstruction, the North-American poles are situated consistently west of the Spitsbergen one. However, after having applied the above-mentioned reconstruction there no longer seems to be any discrepancy. This indicates that the opening-up of the Neo-Arctic basin is a Tertiary event as suggested.

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